Machines for Babbage's Hall

Every machine considered, ranked by interest · 20 October 2026

Every non-electronic computing machine worth considering for the hall, from the list we began with, the three papers (*Early Computing Machines*, *Computing after Babbage*, *Ball-and-disk integrator*) and the rest of the field. Clocks are left out (they are a collection of their own), and so are machines that compute with relays, valves or flowing water (see the end).

Each is scored from 1 to 5 for **interest** (how much there is to see and understand: surprise, motion, history) and for **complexity** (how hard to simulate in bpp and to print and build). The list ranks them by interest as I judge it, the most interesting first; the scores group them, and the order within a score is the judgement. The hall's bays follow the same order, walked the other way: the least interesting at the entrance, the Difference Engine at the far end.

Status: *built*: in the hall, with its STL files; *bay*: has a covered bay "in preparation"; blank: not yet planned.

The ranked list

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
1	Babbage's Difference Engine	1822 / 1849	tabulates a polynomial by adding its differences, with no multiplying	two orders of difference, three digits, is a good model: figure wheels and Babbage's carry, after the Pascaline	5	5	bay
2	Antikythera mechanism	c. 100 BCE	a gear train for the Sun, Moon and planets, the lunar phase, eclipse dials; a pin-and-slot gear for the Moon's varying speed	gears, as in the Differential (convex-hull teeth), and many of them; printed replicas exist	5	4	
3	Enigma (its rotors' stepping)	1923	a keypress steps the rotors by pawls and notches; the current's path through them lights the cipher letter	the stepping is mechanical (the middle rotor's double step comes of itself); the wiring is a lookup in Lua; prints, with wire, contacts and LEDs	5	3	
4	Differential Analyser	1931	wheel-and-disc integrators, wired in a loop, solve differential equations and draw the answer	rolling contact that must not slip; Bush's torque amplifiers can be motors; Hartree built one in Meccano	5	4	bay
5	Zuse Z1	1938	the first freely programmable computer, binary floating point, all mechanical: sliding steel plates and pins	its plate logic is lovely to simulate gate by gate; the whole machine is far beyond us	5	5	
6	Norden bombsight	1930s	a gyro-steadied sight whose analogue computer (rate discs, cams) solves the bombing problem	the computing (disc integrators and cams) is doable; the gyro too, as in Euler's Gallery; whole, very complex	5	5	
7	Ford Rangekeeper Mk I	1917	a battleship's fire-control computer: integrators, differentials and cams solve where the target will be	its parts each are doable (the Differential, integrators, cams); together, a big job	5	5	
8	Curta	1948	a pocket four-function calculator: one stepped drum, nines' complements for subtraction	small and dense: the stepped drum and carry are doable, the whole machine very hard; best as an oversized cutaway	5	4	

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
9	Kelvin's tide predictor	1872	adds ten or more sine waves (the Sun's and Moon's tides) and draws a year of tides	cranks and slotted yokes are easy; the cord over the pulleys that adds them is the hard part (a chain of links, or a computed cord)	5	3	bay
10	Digi-Comp II	1968	marbles rolling down a board of flip-flops count, add and multiply in binary	marbles and levers: contact only, like the dominoes; tuning the flip-flops is the work	5	3	bay
11	Domino Half Adder	modern	dominoes falling add two bits	built: 268 dominoes on printed tiles	4	2	built
12	Pascaline	1642	adds, with Pascal's falling-weight carry	built: three digits, sautoirs, detents, stylus and stop	4	3	built
13	Galton board	1873	balls falling through pins pile up into the normal curve	balls and pins only; easy, and a good opener	4	1	
14	Planimeter (Amsler's polar)	1856	trace a shape's outline and a rolling wheel reads its area	one wheel rolling and slipping on paper; small and clear	4	2	
15	Ball-and-disk integrator	1876	James Thomson's: a ball between a turning disc and a cylinder; the cylinder turns as the integral	the ball must roll without slipping; the integrator the tide predictor's brother and the analyser's ancestor	4	2	
16	Integraph	1880s	follows a curve and draws its integral	a wheel steering by its knife edge; a small step on from the planimeter	4	3	
17	Kelvin's harmonic analyser	1878	integrators find the Fourier coefficients of a measured curve (the tide predictor in reverse)	ball-and-disk integrators in a row	4	3	
18	Thomas Arithmometer	1820	the first mass-produced calculator: stepped drums, a sliding carriage	a stepped drum per digit and a carry; a fine follow-on to the Pascaline	4	3	
19	Leibniz's stepped reckoner	1694	the first four-function calculator: the stepped drum	the stepped drum is the heart of the next three machines	4	3	
20	Odhner pinwheel calculator (Brunsviga, Facit)	1873	pinwheels with retractable pins set the figures; turn the crank	flat pinwheels print well; the carry is the work	4	3	
21	Automatic Totalisator	1913	racetrack betting: a great shaft adder of differentials counts tickets and sets the odds	differentials (built) chained; the first real-time computer, the Science Museum says	4	3	
22	Torres Quevedo's algebraic machines	1890s	find the roots of polynomials with his "endless spindle" logarithmic gearing	special gearing; little known and striking	4	4	
23	Kelvin's equation solver (and Wilbur's)	1878 / 1936	tilting plates on pulleys and cords solve simultaneous linear equations	plates, pulleys, cords: the cords are the hard part	4	3	
24	Michelson and Stratton's harmonic analyser	1898	80 springs sum and analyse waves; it showed the Gibbs overshoot	springs and levers: Bullet springs, many of them	4	3	
25	Comptometer	1887	key-driven adder: press a key and the column moves at once	keys, sectors and carry; no crank	4	3	
26	Digi-Comp I	1963	plastic sliders and pegs: three flip-flops and programmable logic	slides, pegs and a lever: simple parts, many of them	4	2	
27	Dr. Nim	1965	a marble machine that plays Nim, and wins	a few flip-flops and marbles: a little cousin of the Digi-Comp II	4	2	
28	Marble adders (Turing Tumble, Wandel's)	modern	marbles flip levers to add binary numbers	contact and gravity; very like the Digi-Comp II	4	2	

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29	Torpedo Data Computer	1940s	a submarine's analogue fire control	like the rangekeeper, smaller; partly electrical	4	5	
30	The Differential	1827	gears that add and subtract (Pecqueur's patent; older in clocks)	built: printed bevel gears	3	2	built
31	Lehmer's number sieve	1926 / 1932	factors numbers: loops of different lengths (bicycle chains, later toothed wheels) turn until gaps line up	simple shafts and wheels; the 1932 wheel version is easier than chains	3	2	bay
32	Schickard's calculating clock	1623	the first known calculator: adding wheels and Napier's bones	a one-tooth carry gear (it jams on long carries: worth showing beside the Pascaline)	3	3	
33	Jacquard's card reader	1804	punched cards choose which hooks lift: the first stored program	needles, cards and hooks; a reader for a programmable exhibit	3	3	
34	Millionaire	1893	the first direct multiplier: a "multiplication body" of tongues gives each product at once	complex, dense	3	5	
35	Scheutz difference engine	1853	the first working difference engine, with a printer	as Babbage's; one of them is enough	3	5	
36	Three-dimensional cams	1930s	a cam shaped by two inputs gives a function of both (fire control's tables)	one shaped cam and a follower; prints well	3	2	
37	Linkage function generators	1900s	bar linkages that compute a product, a square, a log	a few hinged bars each; a set of them would make one bay	3	2	
38	Slide adder (troncet, Addiator)	1889 / 1920	slides moved with a stylus add and carry by a hook	flat slides; the carry hook is the trick	3	2	
39	Napier's bones, Genaille–Lucas rulers	1617 / 1891	rods laid side by side give a product to read off	no moving mechanism: printable, nothing to simulate	2	1	
40	Slide rule	1620s	multiplies by adding logarithms	one slide; nothing to simulate	2	1	
41	Abacus, suanpan	ancient	beads	nothing to simulate	1	1	

By kind

- **Adding and counting** (digital): Schickard, the Pascaline, Leibniz, the Arithmometer, the pinwheels, the Comptometer, the Millionaire, the Curta, the slide adder, the Totalisator.
- **Tabulating and programmed:** the Difference Engines (Babbage's, Scheutz's), Jacquard's reader, the Zuse Z1.
- **Logic and play:** the Digi-Comps, Dr. Nim, marble adders, the domino adder, the Galton board.
- **Number theory and ciphers:** Lehmer's sieve, the Enigma.
- **Calculus and waves** (analogue): the planimeter, the integraph, the ball-and-disk integrator, Kelvin's tide predictor and harmonic analyser, Michelson and Stratton's analyser, the Differential Analyser.
- **Algebra:** Kelvin's and Wilbur's equation solvers, Torres Quevedo's root finders, linkage function generators, three-dimensional cams.
- **Aiming:** the rangekeeper, the Norden bombsight, the Torpedo Data Computer.
- **Astronomy:** the Antikythera mechanism.

Left out, and why

- **Electromechanical and electronic:** Stibitz's Model K (relays), the bombes, the Harvard Mark I, Colossus, ENIAC and after, and the Mallock machine (electrical). The Enigma is kept because its stepping is mechanical and its wiring is a fixed lookup; the bombe is not, being relays and motors.
- **Water:** Lukyanov's water integrator and Phillips's MONIAC compute with flowing water, which Bullet does not do (rigid and soft bodies only).
- **Clocks and orreries:** a collection of their own.

How the hall uses this list

The hall's bays are walked in rising interest, so a new exhibit goes in its place by its interest score. With eight bays, the hall holds the three built and the five with bays; the Enigma would make a ninth (a fifth row of bays, one spare). The best candidates for new bays, for interest against effort, are the Galton board and the planimeter (quick and clear), and the Enigma and the Antikythera mechanism (striking, and within reach).